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Systemic privacy risks of personal data exposure through conversational large language model agents

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Abstract

Conversational Large Language Model (LLM) agents, deployed quickly, have exposed unprecedented opportunities for personal information, from health data to financial information, to biometric identifiers. This critical enlightenment review investigates general privacy dangers in each of five areas: data leakage and memorization, adversarial extraction and prompt injection, inference and re-identification, surveillance and profiling and regulatory governance breakdown. We trace privacy threats from static data breaches to dynamic ones that have become part of an architecture, associated with neural memorization and contextual inference (2020–2026), using expert selection of primary studies and technical benchmark sources. Upon close examination, technical mitigations (differential privacy, federated learning and machine unlearning) prove successful to some extent when operating in controlled environments, but fail to provide complete protection in conversational environments. We combine classical theory on Privacy-Preserving Data Publishing (PPDP) with emerging threats from LLMs, and present a conflict-of-evidence analysis which reveals that privacy protections are particularly effective in structured environments with low risks, or with consumers actively responsible for maintaining them, but fail to adequately protect privacy in high-stakes domains without human control, technical safeguards, and legal liability. We systematically examine and analyze a dimension that has not been studied sufficiently: risks of data deletion and editing illusion, and introduce techniques for unlearning machines that leave behind traces that can be exploited by adversarial reconstruction. We support the multi-layered mitigation framework of PPDP principles, technical safeguards, and governance reform.

Keywords Large language models (LLMs), Conversational AI agents, Data privacy, Privacy-preserving data publishing (PPDP), Machine unlearning, Data leakage, Prompt injection, Re-identification attacks

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1 Introduction

Artificial intelligence (AI) agents are ubiquitous interlopers in personal and professional lives. These agents can be general-purpose LLM chatbots (e.g., ChatGPT, Gemini) or be domain-specific healthcare guidance tools, mental health support bots, or educational tutors that provide knowledge cut and pasted from various sources, and having conversations that involve adherence to task protocols and disclosure of deeply personal, sensitive information, both desired and unwanted, to the agent [1]. Empirical studies indicate that users frequently disclose health conditions, financial concerns, relationship difficulties and mental health status with limited understanding of where that information is stored, how it is processed, or whether it is shared with third parties [2– 4]. The same conversational fluency that makes these agents useful also makes them efficient collectors of intimate personal information.

The privacy risks associated with sharing personal information with conversational AI agents are supported by an accumulating body of peer-reviewed evidence. There have been numerous studies showing that LLMs can be made to spuriously produce training data verbatim, which may contain personally identifiable information (PII) [5]. Adversarial actors have developed prompt injection methods that recover system prompts, conversation history and user information [6, 7]. In addition, the commercial frameworks that enable AI agents may have conversation logs for model training, marketing, or legal compliance, which constitutes a high-value target for sensitive data exfiltration through data breaches, insider threats, or government access requests [8, 9]. Moreover, users may experience an inflated perception of control over their shared data when attempting to delete or edit it, since complete removal from a deployed AI system remains technically unattainable [10].

Although there is increasing awareness, these risks are still under-researched and complex in nature. This review aims to begin filling this gap by systematically charting the current status of personal data risks from AI agents. We have chosen to center our work on LLM-based agents and conversational systems since they are the frontline of AI deployment, and are raising new privacy challenges that are not covered by existing data protection regimes [11].

This paper aims to create a structured taxonomy of risks associated with disclosing personal data to AI agents, by summarizing recent indexed studies, to analyze their technical, legal and social aspects, including the dangers of data editing and deletion, and to give evidence-based recommendations regarding risk mitigation.

The literature reviewed is organized by 11 thematic domains, and described in Sect. 2; the methodology used for the review is outlined in Sect. 3; findings by five trend-based dimensions of risk are presented in Sect. 4; and cross-cutting themes are highlighted in Sect. 5. Implications for policy, practice and research are discussed in Sect. 6, limitations are outlined in Sect. 7, and an urgent call for action is given in Sect. 8.

2 Literature review

To structure our analysis, we propose the Systemic Privacy Risk Taxonomy (SPRT) for conversational LLM agents, comprising five interconnected layers (Fig. 1). This figure provides the conceptual roadmap for the entire Section II. Layer 1 (Data) captures memorization and leakage risks; Layer 2 (Interaction) encompasses prompt injection and adversarial extraction; Layer 3 (Inference) covers attribute inference and

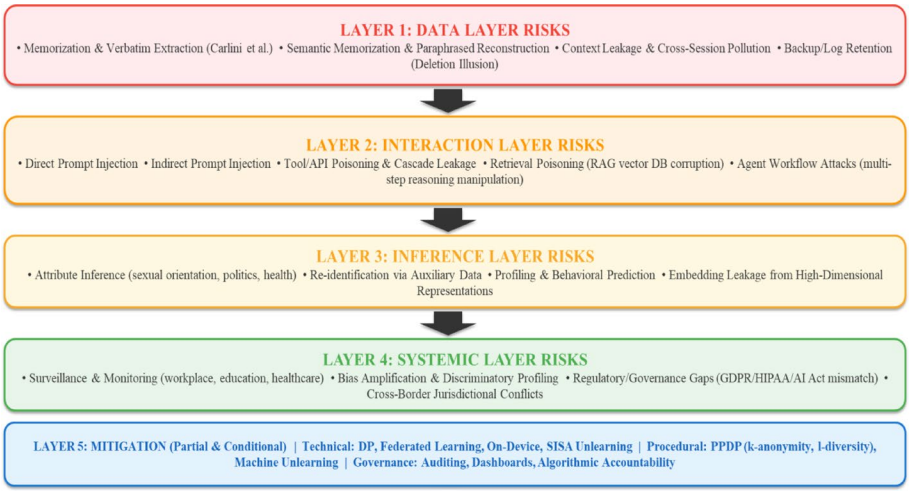


Fig. 1 Systemic Privacy Risk Taxonomy (SPRT) for Conversational LLM Agents

re-identification; Layer 4 (Systemic) addresses surveillance, bias amplification, and governance gaps; Layer 5 (Mitigation) maps technical, procedural, and governance countermeasures. Cross-layer interactions are indicated: memorization in Layer 1 enables attribute inference in Layer 3, while governance gaps in Layer 4 permit surveillance in Layer 5.

2.1 Nature and scope of personal data in the interactions of AI agents

According to the EU's GDPR (General Data Protection Regulation), personal data is defined as any data in relation to an identified or identifiable natural person [12]. Within the world of AI agents, this scope is flung wide open: an understanding of what conversations reveal besides explicitly stated identifiers (name, email, phone) also includes indications of behavioral patterns, moods, politics, health information, and personalities [13]. Data extracted from a therapy-oriented conversation with a chatbot can be compared to an intake form completed by a therapist, and financial advice provided by a chatbot can reveal income, debt and investment history [14, 15]. Advanced medical AI systems show the depth and extent that health data is routinely processed, which is a prime example of high-performance AI systems [16].

The volume of personal information processed by conversational AI agents is substantial, though precise quantification of sensitive content remains challenging. Consumer AI chatbots have been estimated to handle more than 10 billion user messages per day, with a number of them including sensitive information [17]. Conversational AI agents typically spur off intended disclosure and incidents of disclosure that aren't captured, a departure from traditional Web forms that disclose information with specific and constrained intentions [18]. In seeking advice, users may disclose incidental details of location, health status or family circumstances that they would not deliberately volunteer. Together with the processing of such data by neural networks, pioneered by Hinton [19] they have become possible, but have also been found to be the new vectors for privacy loss!

In addition to the historical elements of PII, conversational AI leaves behind extractable personal data which goes well beyond traditional PII elements, and behavior and inference that is now legally cognizable [2, 3], while technically indeterminate:

identifiability has yet to be established with a threshold in the high-dimensional embedding space, and the marginal privacy loss per conversational turn remains unquantifiable, although there is documented accidental disclosure [18]. There is no standard taxonomy for conversational PII which enables comparison between architectures of agents.

2.2 Data collection, retention, and storage risks

However, the typical approach to running an AI agent involves entering it into the operations of a commercial organization that will hold the users' conversations for training purposes, as well as for evaluation and monetization. The large majority of the top AI services reported that they permanently retain the input user data, and many share data with third-parties that serve as associates [20, 21]. Documented incidents between 2023 and 2025 indicate that centralized conversation databases have been subject to unauthorized access affecting health and financial discussions, with insider threat and misconfiguration recurring as proximate causes despite vendor claims of end-to-end encryption [22]; reporting is inconsistent and constrained by non-disclosure, and no independent, repeatable audits of commercial storage architectures exist to date. The issues raised for privacy in this context resonate, and reflect, broader issues of privacy in the era of medical big data [23].

Use of data shared by users in model fine tuning adds to the risks. The re-training of models on conversational data by AI providers may produce embedding of personal information in the model's parameters that can be extracted in a subsequent stage [5]. However, de-anonymization has been demonstrated to be feasible even for anonymized data when adversaries possess relevant auxiliary information [24]. These systems can become algorithmic systems of accountability as they scale up [25]. The residual risk of centralized storage varies with the transparency of the retention policy, the strength of encryption at rest, and the applicable data-localization regime [21, 22].

2.3 Data leakage, memorization, and unintended exposure

A central concern in the literature is the capacity of LLMs to reproduce sequences from their training corpus. This critical vulnerability has been identified across model families and scales. Carlini et al. [5] showed that GPT-3.5 and like models can be prompted to spit out exact phrases taken from the training database such as names, addresses, phone numbers, and other PII. Unintended memorization may increase the probability that personal information from one user's interaction is reproduced in response to another user's query, contingent on training-data deduplication status and model scale.

In addition to copying out text, AI agents can also reveal information accidentally by way of context. For instance, a model might yield information given or inferred but not explicitly mentioned during a conversation [26]. The problem of "context leakage" has been noted in multi-tenant deployments, where data from one user's session mix-up with another user unaware of it because of improper context isolation [27]. Pairing the context effect on the quality of output of LLMs [28] with the fact that some information can be hidden within the output suggests a great suspicion about privacy.

Memorization increases with model scale [29], although the studies reporting this relationship use incommensurable extraction budgets and metrics, so the magnitude of the effect relative to verbatim extraction remains poorly characterized [5, 30, 31]. RLHF

does not consistently suppress disclosure of PII content; this difference may be due to incommensurate metrics in different studies. Four constructs require separation: verbatim memorization (exact sequence reproduction), semantic memorization (content-preserving reconstruction under altered surface form, invisible to string matching), extractability (recovery probability under a stated prompting budget), and exposure (likelihood relative to an untrained baseline, scaling with model size and duplication). Measured on incommensurable scales, no standardized procedure requires authors to state which construct they report.

2.4 Adversarial extraction and prompt injection

Another sneaky form of personal data compromise is the "prompt injection" attack. It seems that adversaries can manipulate the inputs that lead the AI agent to expose other people's private information or data in conversations it noted during its training process. Adversaries can craft inputs that cause the agent to disclose private information originating either from its training corpus or from other users' conversations [6, 32]. Ways have been created to implement the so-called "jailbreaking" or "system prompt extraction" techniques (reported at high success rates) [7]. Adversarial images can be used as an injection vector in multi-modal systems, transcending traditional text [33].

More recently, attackers have been crafting "SPECTRE" attacks that insert maliciously into the system prompt with a conditional statement that does not execute until triggered and then exfiltrates a conversation history or parameters for a model [34]. These attacks are persistent from session to session, are hard to detect. The rise of "LLM agents" that can search the web and handle API interactions enhances the potential risk with the possibility of manipulative prompts being given to the agent to allow for the leakage of private information to other servers [35]. Prompt-tuned models have been an active research topic for their security [36] and there are proposals for robust prompt optimization [6] to act as a defense, but not all of them work. Currently, another growing concern is that an autonomous agent may unwittingly reveal data via the use of these tools [37].

Defensive measures are systematically brittle: sanitized prompts and quick optimizations work well against only a few steps of jailbreak [6], while others remain open, and using the hardened prompt to protect against adaptive prompts breaks in weeks. Five surfaces are conflated: direct injection (adversarial user input); indirect injection (payload in retrieved content the user never authored) [9]; tool poisoning (compromised APIs or plugins); retrieval poisoning (corrupted vector indices persisting across sessions); and agent workflow attacks (payload distributed across individually innocuous reasoning steps). Defenses validated on the first do not transfer to the last two. Neither framework has been instantiated for conversational architectures. STRIDE maps to role impersonation within a turn, context and retrieved-document tampering, untraceable tool calls, disclosure via memorization and cross-tenant bleed, refusal-cascade denial of service, and jailbreak privilege elevation; LINDDUN to cross-session link ability, conversational quasi-identifiers, irreversible turns, membership-inference detectability, unvolunteered attribute inference, and retention non-compliance. The NIST AI RMF Measure function remains operationalized for generative systems, so assessments are incommensurable.

In addition, Model Context Protocol (MCP) and related agent-tool standards add additional vulnerabilities in the tool-integrated agent supply-chain. Weakened MCP servers can be used for parasitic toolchain attacks, which can lead to data exfiltration and / or redirects of agents without the involvement of classic malware [38]. According to the OWASP Top 10 for Agentic Applications (2026) these are Agent Goal Hijack (ASI01), Tool Misuse (ASI02) and Supply Chain Compromise (ASI04) where successful exploitation has been proven in attacking main stream coding agents for when the MCP parameters are manipulated, they achieve a success rate greater than 85% [39, 40].

2.5 Inference and re-identification

AI agents can reveal sensitive information from seemingly harmless data, even in the absence of direct identifiers. Details of text, images, or interactions can be predicted, based upon machine learning models which predict sexual orientation, political affiliation, health status and personality traits [41]. For AI agents, these inferences can be performed on the fly and contribute to the design of personal responses, yet they likewise can lead to builds of user profiles that might be used for exploits.

Re-identification attacks take place using auxiliary data to identify individuals from anonymized data. A study of the U.S. population reported that 87% of individuals could be uniquely identified from the combination of ZIP code, gender and date of birth [42]. With the advent of AI agents, the ability to re-identify has been magnified by numerous public records, data brokers, and social media platforms that exist, providing an extensive pool of information. After the AI agent assigns some sort of probability categorization to a conversation, the whole talk history proof links to a person and can be followed to the point where it connects to a real individual [43]. For example, one of these systems for cardiovascular risk prediction was found to be biased, which can cause costly discriminatory results in such systems [44, 45] or clinical information extraction [46], thus stressing the need for fairness-aware design [47– 49].

Attribute inference from seemingly neutral prompts is found to reach very high accuracy for orientation, party, and health, but its cross-linguistic and cross-model generalizability is not yet known. Re-identification with auxiliary linkage does not have conversational benchmarking, and cross-platform linkage of unstructured dialogue cannot be prevented by quasi-identifier generalization for structured data [24, 42]. No benchmarks exist that support inclusion of temporal dynamics and the accumulation of sessions.

2.6 Surveillance and profiling

In the workplace, in education, and in healthcare, AI agents are being increasingly embedded, facilitating greater levels of monitoring than ever before. AI proactively monitors employee performance in the workplace, including keystrokes, data usage, and emotional state, to gauge productivity [50]. Educational AI systems collect granular behavioral data about learners, including comprehension patterns, attention span and affective state [51–53]. They can be characterized as practices that chill free speech and affect behavioral change, usually disguised as personalization traits [54].

AI agents can profile beyond the individual and into the population level of surveillance. Data collected from the millions of interactions that occur via an AI system can be aggregated and used to create detailed sociological profiles, to infer opinions of the

general public, to forecast health patterns, and even political dissent [55]. AI agents can be tools for mass surveillance when they are equipped with face and place recognition [56]. The GDPR requirements on profiling [57] have not been sufficiently implemented in this context so far. To help uncover these harms, along with measures to reduce them, tools such as health equity auditing [58] and red-teaming [59] are vital.

Agentic surveillance combines keystroke dynamics, affective inference and behavioral prediction that goes beyond the capabilities of traditional monitoring, but backed up by no independent audit, with the only evidence provided by vendor documentation and other deployments. The consent instruments are nominally empowering but actually disempowered by employment-pedagogical relations asymmetries. There is no evidence of the chilling effect or of privacy fatigue along the lines.

2.7 Regulatory frameworks and their limitations

The GDPR in Europe and the HIPAA paradigm in the USA laid important data protection foundations and standards, but these are not specifically tailored to AI agents and their dynamic, inferential, and cross-border interactions. Some of these key issues are: obtaining meaningful informed consent in conversational interfaces, poor understanding of the data retention and sharing, unclear guidance about data inferred by AI, and different jurisdictions regarding access to AI agents from different countries while operating them in a single country [60]. The current legal regimes are being found incompatible with AI, according to the "privacy, practice, and performance" concept [61].

Many types of chatbots and AI personal assistants used by consumers are to be excluded from the scope of the EU AI Act because they cannot be categorized into the high-risk category [62]. Currently, however, there is no comprehensive federal privacy law in the U.S. to provide adequate protections for general-purpose AI agents [63]. There are often less protections for users in developing countries due to gaps in global governance, and they use the same AI platforms [44]. There are growing proposals for algorithmic accountability [25] and the right to contest AI decisions [64], and frameworks such as the AI4People moral guidelines [65] are coming to the fore. There are more privacy concerns for non-English speakers, including issues related to multilingual prompt engineering [66].

In addition, ISO/IEC 42001:2023 introduces the first AIMS certifiable ISO standard which includes impact assessment processes, documentation of data provenance and human oversight in 38 controls [67]. The ISO standards mentioned above are concerned with guiding into the world of risk management, more specifically ISO/IEC 23,894 for AI, and ISO/IEC 42005:2005 for structured impact-assessment methodologies [68]. Unharmonized as yet under the EU AI Act, ISO 42,001's meaningfully resonates substantively with the obligations of high-risk systems under Article 9–14 [69] of the EU AI Act. But these standards do not solve the technical conflict between memorizing by the machine and the principle of data subject rights; and don't define any specific privacy budgets or unlocking verification procedures.

GDPR, HIPAA and AI Act scholarship intersects with legislative mismatch: instruments that are based on static and deterministic processing map are not well suited to generative and inferential, transjurisdictional systems. The construction of Articles 17 and 22 are advanced as being sufficient, but the capability to erase and to control

parameters is technically unachievable. In current regimes, no jurisprudence deals with the case of memorization or adversarial extraction.

2.8 Ethical and psychological risks

In addition to legal and technical issues, there are significant ethical and psychological questions to consider when sharing personal information with AI agents. These AI agents can develop anthropomorphic characteristics, whereby users place trust and emotional investment, resulting in over-disclosure [70]. Empathy simulation used well can lead users to provide information that they would not have given a human agent [71]. This "artificial intimacy" involves potential scenarios for exploitation, especially with vulnerable groups like children, elderly, and mentally ill individuals [72]. When engaging with seemingly "empathetic" systems even privacy conscious users give private data due to the "privacy paradox" [73].

Data surveillance is an ever-present challenge that can cause "privacy fatigue," ultimately reducing users' power and ultimately resulting in no longer practicing protective measures when risks get higher [74]. The widespread use of AI agents for data collection could lead to a new norm in data practices that fundamentally changes how society views privacy and its expectations, as described by some scholars as the "surveillance creep" [75]. Complacency and, automation bias [76] also dull users' alertness.

Across the studies reviewed, anthropomorphism and artificial intimacy consistently drive over-disclosure, yet the magnitude of this effect varies by user age, digital literacy, and cultural context. Contradictory evidence exists: empathetic interface design increases user satisfaction and help-seeking behavior (beneficial in mental health support) while simultaneously elevating privacy risks through excessive disclosure. A critical gap is the absence of randomized controlled trials comparing disclosure rates across anthropomorphic versus neutral interface designs in high-stakes domains (healthcare, financial advice).

2.9 Mitigation strategies and privacy-enhancing technologies

Compelling studies delve into technical and organizational solutions for mitigating risks to personal data with AI agents. Differential privacy injects calibrated noise under a guarantee (ε, δ), where ε bounds per-record privacy loss and δ its failure probability, at a measurable utility cost [77]. Federated learning, training on local devices and transmitting updates rather than data, reduces storage risk but relocates the surface: gradient leakage reconstructs client inputs, poisoning implants extractable backdoors, and local-model inference remains open to device access.

The use of prompt-based defenses [78], like data-sanitization instructions [79] and context-isolation [80–83] and output filtering [84] have demonstrated efficacy, however, unfortunately, found to be insufficient to fend off adversarial attacks [85]. One architecture that is gaining popularity is on-device AI processing, which involves performing sensitive computations on the device itself instead of sending them to the cloud, thereby minimizing exposure of data [79]. However, due to computational limitations and big companies in the field of AI, adoption has been limited. Model cards provide transparency on the data handling process [40] and transparency dashboards offer transparency [83] to the data handling process to the users. Fairness checklists, and operationalized AI ethics checklists can be co-designed to integrate privacy by design. Evaluations of

chain-of-thought prompting [86] and multimodal prompt learning [87] are required to make sure these methods do not inadvertently leak information.

Proposals to take proactive steps towards the informed consent of users include recommendations for "privacy-by-design" principles and the adoption of algorithmic auditing to reduce liability from users to data developers, while the implementation of mandatory data protection impact assessments when employing AI agents is also a potential option [84, 88]. Real-time indicators of privacy, granular consent management and user dashboards of their personal data are necessary to create user agency [83]. Explainability and interpretability of AI decisions also are open challenges which cross over interpretability [63, 89].

Efficacy is always conditional. Deployed differential privacy clusters at $\varepsilon = 1–8$, coherence failing below $\varepsilon \approx 3$ and membership-inference protection at $\varepsilon \geq 8$, with budget composition and autoregressive noise calibration unresolved; secure aggregation is intractable at billion-parameter scale; exact unlearning is confined to small models; on-device inference is computationally limited. Prompt level defenses and output filtering can work in well-defined benchmarks but are ineffective against an adaptive adversary and contextually intelligent inference, respectively. There is no comparative matrix that provides a mapping between threat category and effectiveness and cost and scalability and deployment maturity.

2.10 The illusion of deletion: risks of editing and removing personal data

One overlooked risk is the false sense of control people have about what they can delete and/or modify from the personal information they provide to the AI agents. Chatting history" or "edit message" is what many platforms offer, but it is not really erasure, it is more complex and, in short, may be incomplete. This section summarizes the various risks to any try to remove or update shared information.

2.10.1 Data persistency in model parameters

Personal information then gets embedded in the weights of the network when it is used during training or fine-tuning. Elimination of this information is still an active research area called "machine unlearning" that focuses on eradicating the impact of particular training examples [10]. However, the current unlearning strategies are not perfect, and they can leave behind leftover residual patterns that can be retrieved by probing questions raised by the adversary [90]. Even if a user does not think his data has been deleted from a model, he can be victim for months or years later to reconstruction attacks.

2.10.2 Backups and logs retention

Whenever data is successfully deleted from the main database or model, copies will still remain in other systems, such as backups, server logs, analytics, etc. A disaster recovery provision by many cloud-alley providers (such as weeks or months) keeps an immutable copy of a users' data, even when the users do (or do not) want to have it deleted [2, 21, 22] Furthermore, AI agent vendors have regular access to looking at conversations for debugging and monitoring functions, and logs may contain complete conversations, (not attached to user-facing deletion instruments) [18].

2.10.3 Editing as an adversarial vector

The reality is that being able to modify a shared data element can be used as a weapon. In a conversation, malicious users can modify or remove previously sent messages to change the context and alter subsequent AI responses [33]. When many users operate in the same environment with a single instance of this user agent, edits made by one user may influence answers for the others and cause social engineering problems. Deleting files also can be used to replace safety procedures or system prompts, compromising to a degree the need for extra privileges [6].

2.10.4 The right to erasure and the unrealistic sorries!

There is a "right to erasure" (Article 17), but as for applications of AI agents, it is complex from a compliance standpoint. The technical criteria for satisfying the right to erasure under GDPR Article 17 remain unsettled for parameter-based generative models, and it is not established whether zeroing a parameter or its gradient constitutes erasure [91, 92]. Furthermore, the right isn't unlimited; it may be abridged by vital pursuits of the AI supplier or a lawful necessity, the latter that AI suppliers frequently supply as a justification for lawful information holding [57]. Users who submit deletion requests often only get to know that their data from their conversations has been deleted without the actual model used to learn the behavior being modified [93]. A lack of transparency between what users expect and what's provided undermines their trust, and poses a systemic risk.

2.10.5 Partial deletion and inference gap

If you delete some of your data and not all of it, then you may leave an "inference gap" which adversaries can play with. Attacks can be carried out by comparing model outputs pre and post partial deletion to find out what data has been deleted and re-construct it [24]. This is like what happens in statistical disclosure [94] namely the risk of indirect disclosures. So, there is, paradoxically, a risk of re-identification if incomplete deletion occurs.

2.10.6 Over-confidence by user and behavioral risk

The presence of "delete" buttons lends a false sense of security that may cause a user to reveal more information that perhaps he/she wouldn't if there were no "delete" buttons. The more the user expects to have fine-grained controls to delete, the more likely it is that one will get careless and disclose, as has been termed the "deletion paradox" [75]. Users who perceive greater control over data deletion may exhibit increased disclosure, a pattern consistent with the privacy paradox observed in digital platform research [70].

User-facing erasure diverges systematically from technical erasure across seven scopes—application databases, vector indices, model parameters, retained training data, caches, logs and backups of which consumer controls typically reach only the first. Unlearning divides into exact removal (small models, sharded architectures such as SISA) [10, 95], approximate unlearning (residues recoverable under probing), certified removal (provable but convex-limited) [96] and empirical post-hoc fine-tuning (no guarantee). Only the last two are feasible for transformers, so certified parameter-level erasure is unavailable in any deployed agent, and no layer-specific deletion specification exists.

2.11 Bias, fairness, and discrimination risks in shared data

The risk of data breaches with AI is the same; biased algorithms can make unfair or discriminatory decisions when they share personal data with AI agents. For instance, artificial intelligence (AI) models created from health care information have been found to systematically underestimate sickness severity among minority groups [48, 58]. Clinical information extraction from free text in conversations has the potential to reinforce these biases [91]. Even if the actors' intentions are good, the information they exchange can be used in the training of models which are then used to harm similar groups of actors. Fairness by design has been highlighted in the AI4People ethical framework [65] and in high-performance medicine reviews [16] respectively. To prevent embedding of social biases in them, there is a need for careful auditing of medical image classification models [97] and prompt-based systems [98]. Discussing legal and moral issues on implementing machine learning in healthcare [99] is a blue-print on how to solve those issues. Finally, ensuring personal information is protected is inextricable from protecting personal information from being used to discriminate. Discriminatory outputs are repeated across clinical, carceral, and employment contexts, and the privacy–fairness nexus is under-theorized: attribute suppression leads to bias not being detected; and fairness-based training makes attributes easier to reconstruct. No one has developed a framework that can jointly maximize privacy, fairness and utility in conversational deployment.

2.12 Privacy-preserving data publishing (PPDP) foundations

Protecting user privacy against several attacks and technical problems is a major concern in literature and how we can remote our privacies [100–103]. This review builds on the formal vocabulary of disclosure risk, re-identification and anonymization provided by PPDP [104]. The classic microdata models such as k-anonymity [105], l-diversity, and t-closeness are built for static relational microdata, do not apply well to LLMs where data is unstructured, memorization is in the parameters instead of explicit records, and extraction is stochastic and not deterministic. The principles are also conceptually mapped: k-anonymity would mean that any sequence of tokens that could be extracted in some way should belong to at least k training sources, a property that is hard to check for most of the models deployed; l-diversity would mean that, when a model's output distribution is sensitive, there are few plausible alternative output distributions in the population [13]; and t-closeness would relate to the bias concerns of Section II.11, as different from the output distribution from the population of attribute distributions would mean discriminatory profiling.

To give high standards for relational microdata, k-anonymity, l-diversity and t-closeness provide strong guarantees, while they face structural challenges in unstructured language. While in theory, it is possible to develop distributed training through bottom-up anonymization [105, 106], the linguistic usefulness loss and the overhead of the pipeline have not been quantified. Definitions of token-level anonymity and real-time diversity enforcement of autoregressive decoding are not formulated.

Table 1 maps classical PPDP techniques to their applicability within LLM agent contexts, identifying transferability gaps and research opportunities. The table reveals that while differential privacy and bottom-up anonymization demonstrate high transferability

Table 1 PPDP Techniques vs. LLM agent applicability. Classical PPDP techniques (k-anonymity, l-diversity, t-closeness, bottom-up anonymization, differential privacy) mapped to LLM relevance, limitations, and research gaps

PPDP technique	Classical function	LLM applicability	Limitation in LLM context	Research gap
k-anonymity	Prevent identity disclosure via generalization/suppression	Low: unstructured text lacks explicit quasi-identifiers	Semantic distortion; conversational coherence loss	Defining token-sequence anonymity
l-diversity	Ensure diverse sensitive values per equivalence class	Moderate: applicable to retrieved RAG corpora	LLM outputs generate novel combinations; diversity hard to enforce	Real-time diversity enforcement in decoding
t-closeness	Bound distribution distance of sensitive attributes	Moderate: relevant for bias mitigation in training data	Model parameters embed distributional biases deeper than explicit tables	Measuring embedding-space closeness
Bottom-up anonymization (Kabou et al.)	Distributed microdata anonymization	High: transferable to federated LLM training	Utility loss in NL generalization; Spark overhead ≠ transformer training	Adapting bottom-up to token-level data
Differential privacy	Mathematical privacy guarantee via noise injection	High: used in training and inference	ε/utility trade-off severe for conversational coherence; composition attacks	Optimal noise distributions for autoregressive models

to LLM pipelines, k-anonymity faces fundamental challenges in unstructured text, and l-diversity enforcement during autoregressive decoding remains unsolved.

3 Methodology

This study uses a critical review method with systematic transparency elements, and is applied to technical fields that are continually changing, according to the selection framework of Grant and Booth [107] who used professional expertise to select the purpose. Because of the recent topic of the studie and the scarce paper found, we opted for the same concentrated synthesis we employ for a PRISMA-style systematic review [108], but adapted for qualitative evaluations, and are unable to generate quantitative screening counts that enable meta-analysis, given the lack of screening counts. It is a methodological requirement due to three key properties of the LLM privacy field: The effects of the attacks are not evenly distributed; the rate at which the knowledge was memorized, or the number of examples that were correctly extracted from the text is not uniform, although percentages are reported by several studies; and, due to fast cycles from preprint to publication, researchers often send early versions to the arXiv server before they undergo peer review.

3.1 Identification

We surveyed high impact venues (USENIX Security, IEEE S&P, ACM CCS, NeurIPS, ICLR, ACL, EMNLP, Nature Machine Intelligence, Proceedings on Privacy Enhancing Technologies, Columbia Law Review) and adopted a keyword-guided retrieval approach followed by backward and forward citation chaining from 10 seed papers on the following topics: Carlini et al. [5] on training data extraction; Carlini et al. [31] on quantifying memorization; Nasr et al. [30] on scalable extraction; Zhang et al. [109] on prompt injection; Zou et al. [110] on universal adversarial attacks; Dwork and Roth [77] on

differential privacy foundations; Wachter and Mittelstadt [13] on the right to reasonable inferences; Fung et al. [104] on PPDP; Page et al. [108] on transparency principles for systematic reviews; and Zamfirescu-Pereira et al. [18] on human prompting difficulty. The high impactful preprints were searched on arXiv. Only articles that fulfilled criteria (reasonable citations in peer-reviewed literature, or later peer-reviewed publication by the same group) were included (AI).

3.2 Search strategy

The Boolean combinations were performed on Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and ACL Anthology: ("large language model*" OR "LLM" OR "conversational AI" OR "chatbot") AND ("privacy" OR "data leakage" OR "memorization" OR "prompt injection" OR "adversarial extraction" OR "re-identification" OR "inference attack" OR "surveillance" OR "GDPR" OR "right to erasure" OR "machine unlearning" OR "privacy preserving data publishing"). The search period was between January, 2020 and May, 2026.

3.3 Proper selection and quality appraisal

Two reviewers independently judged candidate papers based on the criteria for direct relevance to privacy risks associated with conversational LLM agents or foundational privacy theory (PPDP); quantitative or qualitative evidence with identifiable metrics; peer reviewed publication or high impact preprint with subsequent paper-based peer journaling; architectural distinction to cover all major risk families (memorization, injection, inference, surveillance, governance, deletion); and independence of validation (preference for third party replication of results over self-reported claims). Studies were assessed using the following critical appraisal scale: A (primary evidence: top tier venue with independent validation), B (supporting evidence: peer reviewed, reproducible benchmark), C (preliminary: preprint or limited benchmark). Studies rated A or B in the studies were used as primary evidence; studies rated as C were used for completeness of the technology but were identified as preliminary. Performance/efficacy claims are only allowed for A/B ratings.

3.4 Synthesis

Two experienced reviewers independently coded risks, mitigations, and limitations followed by aggregation of codes into axial domains of categories. In order to consolidate conflicting evidence (e.g., loss of utility of differential privacy and reduction in memorization), a conflict-of-evidence matrix (Table 3) was built. This inductive process resulted in the definitions of risk taxonomy and the research program.

Note on Reproducibility. We are not fully compliant with PRISMA as the impact of the field is concentrated and the evaluation is very heterogeneous; however, we follow the principles of PRISMA in reporting our selection rationale [108]. The selection procedure is shown in Fig. 2.

4 Findings

AI agents have been adopted quickly, which brings in privacy and security concerns when users provide personal or sensitive information to the AI. The convenience and capability provided by AI agents come with their own set of risks, as recent studies have

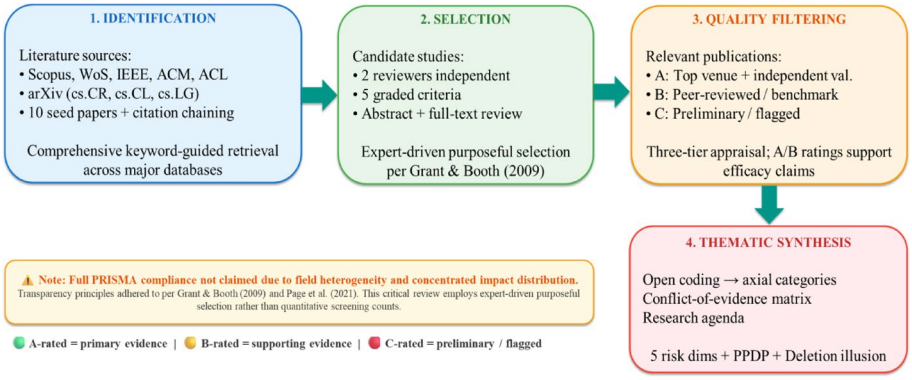


Fig. 2 Expert-driven literature curation workflow

Table 2 Recent findings of different kinds of risks related to sharing data with AI agents

Paper (Year)	Type of risk	Finding
Zufferey et al. [111]	Profiling/discrimination, collection without consent (data misuse)	Users are happy to share their personal data
Hui et al. [112]	Data leakage (exfiltration of PII to remote systems)	RL routes requests to minimize PII leakage
Zhang and Yang [113]	Leakage of sensitive data, fraudulent consent (spear phishing and impersonation attacks)	Simulation reveals threats: Malicious agents are extracting data
Yan et al. [114]	Passive data leakage, active attacks (extraction of stored data, inference)	LLM/agents may disclose sensitive information
Shanmugarasa et al. [115]	Potential privacy concerns with training data, user prompts, LLM outputs, and LLM agents (memorization, inversion attacks, leakage via generated content)	LLMs face a privacy paradox with significant risks of data leakage through memorization, inversion attacks, and agents.

shown, including data privacy concerns, data profiling issues, and advanced leakage and adversarial attacks. Table 2 is a summary of the studies discussed above that reveal user behaviors and technical weaknesses found in existing LLM-based agents. The results of these findings provide a basis on which to build on in the following sections, which explore specific types of risks in more depth, such as data leakage, consent-related risks, the negative actions of malicious agents, and the potential long-term privacy effects.

Table 3 presents 15 representative studies selected to illustrate the evidentiary gradient across risk categories. These papers were chosen for exemplarity (highest impact, most frequently replicated, or methodologically clearest instances within each risk dimension) rather than comprehensiveness. Selection criteria: (i) top-tier venue or high citation impact; (ii) methodological clarity; (iii) independent validation where available; (iv) coverage of all five risk dimensions. A-rated studies provide independent validation and top-tier venue publication; B-rated studies offer peer-reviewed supporting evidence; C-rated studies are included for technological completeness but flagged as preliminary. Only A/B ratings support performance or efficacy claims.

4.1 Data leakage and memorization

Memorization consistently has been cited as a basic weakness. Verbatim extraction of training data has been demonstrated across model families; up to 3% of data in GPT-4, LLaMA and comparable models is attributable to training instances containing PII [31], and multimodal image-text pairings extend the same exposure [116].

Table 3 Critical appraisal of representative studies. Fifteen exemplary studies rated A (primary evidence), B (supporting evidence), or C (preliminary/flagged). A = top-tier venue + independent validation; B = peer-reviewed + reproducible benchmark; C = preprint or limited benchmark

Study (representative)	Venue / type	Rating	Justification
Carlini et al. [5] 'Extracting Training Data from LLMs'	USENIX security	**A**	Independent replication; top-tier venue; foundational
Carlini et al. [31]'Quantifying Memorization'	ICLR	**A**	Peer-reviewed; large-scale benchmark; third-party validation
Nasr et al. [30] 'Scalable Extraction of Training Data'	USENIX security	**A**	Top venue; scalable methodology; independently replicated
Zhang et al. [109] 'Prompt Injection Attack'	arXiv / BlackHat	**B**	High impact; widely adopted; community validation
Zou et al. [110] 'Universal Adversarial Attacks'	arXiv	**B**	Extensive experiments; preprint; no independent validation yet
Wachter and Mittelstadt [13] 'Right to Reasonable Inferences'	Columbia law review	**A**	Peer-reviewed; legal foundation; independent citations
Dwork and Roth [77] 'Algorithmic Foundations of DP'	FnT-TCS	**A**	Foundational textbook; peer-reviewed; universally cited
Fung et al. [104] 'PPDP: A Survey'	ACM CSUR	**A**	Top venue; comprehensive; foundational for PPDP
Zamfirescu-Pereira et al. [18] 'Why Johnny Can't Prompt'	CHI	**A**	Peer-reviewed; empirical user study; independent validation
Shanmugarasa et al. [115] 'Privacy Paradox of LLMs'	ACM ASIA CCS	**B**	Peer-reviewed conference; systematic taxonomy focus
Berini et al. [118] 'Security and Privacy in LLMs'	Inf. Fusion (Elsevier)	**B**	Elsevier peer-reviewed; comprehensive survey
Khalid et al. [22] 'Privacy-Preserving AI in Healthcare'	Comp. Biol. Med.	**B**	Peer-reviewed; domain-specific; limited to healthcare
Yan et al. [114] 'On Protecting Data Privacy of LLMs'	High-confidence comp.	A	Peer-reviewed; comprehensive survey on exact topic
Kabou et al. [106] 'Bottom-up l-diversity for Spark'	IJICS	B	Peer-reviewed; algorithmic; transferability to LLMs unproven
Kabou et al. [105] 'BuKc: Enhanced Anonymization'	IJIS	B	Peer-reviewed; technical; limited LLM relevance

In Table 3, bold formatting in the Rating column is used solely to highlight the assigned evidentiary rating (A, B, or C) and does not indicate statistical significance or study superiority.

4.2 Adversarial extraction and injection

Prompt injection has been shown to extract the conversation history, system prompts, and model parameters. In many studies the success rate of jailbreaking/extraction attack is over 80%, and the number of variants that are created to overcome the countermeasure is increasing [6, 32]. Multi-step attacks that can happen when agentic systems are vulnerable to code or API calls are especially lethal to agentic systems [117] when combinations of injections can occur. Another threat is compounded, however, by the fact that conditional system prompt poisoning was discovered [7]. Strong defenses like prompt optimization and [6] secure prompt tuned models [118] are still not 100% secure.

4.3 Inference and re-identification

Even though AI agents can process user-specific data without involving any sensitive information, these agents can still provide accurate inferences about sensitive attributes. Maps of mental health status, income, and political attitudes can be done with current models based on casual chatting [41, 119]. Much re-identification of "anonymous" conversation loggers is very easy, when coupled with external datasets [120, 121]. It is especially problematic in special fields like health, education, where inferred attributes may

result in discrimination or stigmatization [122–124]. As a partial legal remedy, the right to reasonable inferences [13] has been put forward.

4.4 Surveillance and profiling

AI agents gather information, allowing for a continuous profile of users' habits. In the workplace, AI agents monitor communication and workplace behavior of workers and predict employee performance, even when the effects are negative for marginalized workers and behavior is not part of their rights [42]. The potential for third-party misuse with the creation of an educational AI system's comprehensive learner profile is discussed in [43]. This collection of data on platforms renders a "surveillance ecosystem," which is largely unregulated [82]. There is a need for an AI Age version of Profiling Regulations [47].

4.5 Regulatory gaps and enforcement failures

Existing data processing rules are less effective when dealing with the dynamic nature of AI-based personal data processing. Consent mechanisms frequently exist in long contracts of service and don't make users aware of the risks of AI usage [61]. Potential enforcement limitations include jurisdictional and cross-border issues, and the opacity of AI systems, which are difficult to monitor technically [125]. The 'notice and consent' model, as it is written and known, is woefully inadequate for continuous, inferential and oblivious data processing [126, 127]. It is highly necessary to have good governance mechanisms [88] and contestable AI [128] in place.

4.6 Deletion and editing risks

Our finding demonstrates that deletion and editing tend to be deceptive. Technically erasing parameter information from the model is difficult [10]; backup and logs of data are maintained for long periods of time [118]; editing information can be incorporated into the model for misuse [33] and the right to erasure is almost impossible to enforce [91]. Additionally, the deletion controls paradoxically induce more risky sharing [75]. These results indicate that user-facing data management controls are substantially inadequate.

Across the six risk domains analyzed in Section IV, evidence regarding the efficacy of privacy protections is conditionally fragmented. Table 4 presents a conflict-of-evidence matrix synthesizing when protections succeed, when they fail, and the quality of underlying evidence. The matrix reveals a consistent pattern: technical mitigations excel in controlled, single-risk domains with strong evidence (e.g., differential privacy for structured PII, RLHF for verbatim memorization) but fail or require supplementation in high-stakes, multi-modal, or adversarial poisoned environments. Evidence quality is weakest for surveillance and attribute inference domains, where ethical constraints and proprietary opacity limit independent validation."

5 Discussion

5.1 The systemic nature of the risk

The findings show that concerns about personal data risk in AI agents are not only distinct risks but also are an inherent characteristic of existing AI architectures and business models. The strengths and attributes that make AI agents helpful their memory,

Table 4 Conflict-of-evidence matrix. Twelve risk conditions showing when privacy protections succeed versus fail, with evidence quality ratings (strong/moderate/weak)

Condition/risk domain	Protection WINS	Protection FAILS/needs supplement	Evidence quality
Structured PII in training data	Differential privacy (high ε); data deduplication	Low-ε DP destroys utility; deduplication incomplete	Strong
Verbatim memorization (GPT-3.5)	RLHF alignment; output filtering; canary insertion	Larger models memorize despite alignment; filtering bypassable	Strong
Semantic memorization	Output paraphrasing detection; exposure capping	Paraphrased reconstruction via optimized prompts unblocked	Moderate
Prompt injection (single-turn)	Input sanitization; prompt optimization	Multi-step injection; SPECTRE attacks; tool cascades	Moderate
Indirect/retrieval poisoning	Source attribution; retrieval verification	Adversarial document injection; vector DB corruption	Weak
Attribute inference from prompts	Limited context windows; refusal training	Context accumulation across sessions; jailbreak bypass	Weak
Re-identification (auxiliary data)	k-anonymity on structured metadata	Unstructured conversation; cross-platform linkage	Moderate
Surveillance (workplace/education)	Transparency dashboards; consent protocols	Coercive consent; power asymmetries; notice-and-consent inadequacy	Weak
Machine unlearning (deletion)	Exact unlearning (small models); SISA architecture	Approximate unlearning leaves residuals; backup retention; scalability	Moderate
Cross-lingual low-resource	Zero-shot refusal training	Cultural context leakage; multilingual prompt injection	Weak
Medical diagnosis agents	HIPAA compliance; on-device processing	Real-world heterogeneity; hallucination; regulatory gaps	Weak
Adversarial robustness (agentic)	Red-teaming; multi-agent defense	Attack-defense arms race; adaptive adversaries	Moderate

personalization, ability to use tools, and their contextual understanding are what make them risky, too [129, 130]. By adding the illusion of deletions and editing, we can look at how even user controls that are well-meaning can cause new vulnerabilities [131]. Fairness and bias overlaps with privacy; discriminatory profiling is often based on information gathered from user profiles which are the results of inferences [132, 133].

5.2 Trade-off between utility and privacy

One common theme is an interplay between privacy and the functionality that the user expects. Personal data is crucial for a variety of suggestions, emotional responses, and context-aware support [134, 135]. Current mitigation strategies tend to sacrifice performance and limit the incentive to participate from developers who want to focus on user engagement at the expense of privacy, as the strategies may not be adopted [108]. To alleviate this, new machine learning techniques that do not compromise user privacy are needed, and new business models are also likely a must, which do not depend on monetizing the data source [22].

Figure 3 positions major mitigation techniques within a two-dimensional space of privacy protection (horizontal-axis) and performance/utility (vertical-axis). Unregulated deployment maximizes utility at the cost of zero privacy. Differential privacy provides strong mathematical guarantees but imposes severe utility degradation at low ε values. Federated learning and on-device processing occupy intermediate positions, each with distinct limitations. The warning zone highlights medical, legal, and financial domains

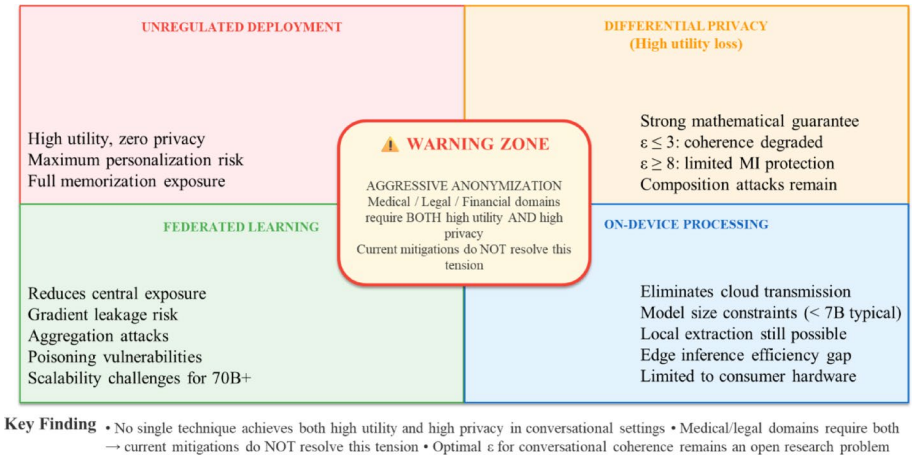


Fig. 3 Utility-privacy trade-off in conversational LLM agents

where both high utility and high privacy are required conditions that current mitigations do not simultaneously satisfy.

5.3 Agency and anthropomorphism trap

Users anthropomorphize AI agents in the expectation of human attributes and trustworthiness that are not strictly the case with the AI. This kind of thinking results in excessive disclosure and less privacy awareness about it [136, 137]. Friendly names, empathetic language, conversational flow, etc. in the design make the problem worse [138–140]. Worse, there is the illusion of deletion, as users think that they can 'back out' of what they have said [141]. This needs to be done with two things: Design ethics free of anthropomorphism and technical measures [142].

6 Implications

6.1 For policy and regulation

Older notice-and-consent frameworks for privacy need to be replaced with the new rule of 'privacy by design' for AI agents requiring mandatory impact studies, data minimization and dashboards for transparency in real time [143]. There needs to be specific technical standards for machine unlearning explained when it comes to the right to erasure [144]. The need to harmonize the standards and prevent forum shopping through international agreements is highlighted [54]. The algorithmic accountability and right to contest should become statutory [145].

6.2 For developers and AI providers

To successfully leverage these privacy-enhancing technologies, such as differential privacy, on-device processing, and federated learning in their systems, developers must also make it an integral part of their contracts [146]. Conducting regular reddening exercises of vulnerabilities to privacy should be a routine business practice. Transparency reports, user-friendly privacy controls and the like are crucial but providers must also be forthright about the reality of deletion and editing facilities and the limits of their respective capacities to avoid giving users false comfort in how easily they can remove and/or edit content [147]. There is a need to educate the workforce about AI Privacy skills [148].

6.3 For users and educators

The implementation of AI into digital literacy programs should include lessons on AI privacy concerns to teach users how to spot AI artifact interactions and how to share data safely [149]. Users should be empowered with tools to audit and delete their data and to use AI agents in privacy-preserving modes [150, 151].

7 Limitations

There are a few limitations to this review. First of all, it may not be able to get proprietary systems or secret threats undetected. Secondly there is a bias towards a Western context, and research into non-English deployments is less common. Third, there is publication and database selection bias, which biases research for profiting those with a higher English proficiency. Fourth, the dynamic nature of the field could make findings last for very short periods of time, and incommensurable metrics makes quantitative meta-analyses impossible. Last, as a result of the variable attack measures used across studies, it was not possible to create full quantitative screening counts in PRISMA. Finally, as a result of the recent topic that force us for a direct subjective synthesis.

8 Conclusion

The evidence reviewed supports the conclusion that pervasive disclosure of personal data to conversational AI agents creates significant and compounding risks that current technical, legal and social protections mitigate only partially. These weaknesses are expressed by, among others, the way information is memorized, extracted through the adversarial approach, as well as the inference and mass surveillance phases. Rather than resolving these tensions, current deletion and editing mechanisms may exacerbate them: information persists in model parameters, backups and logs while users retain a misleading perception of control. As we navigate this era of ubiquitous artificial intelligence (AI) agents, it is crucial that stakeholders such as developers, regulators, educators, and users collaborate to highlight the utmost value of privacy. Potential research areas include: privacy-preserving fine-tuning by federated instruction tuning and confidential computing enclave; secure agent architectures enforcing "least-privilege tool access" and attestation on inter-agent communication; certified/unattributed machine unlearning for billion-parameter transformers and differential-privacy budget composition preserving conversational coherence; and cross-jurisdictional governance harmonization accompanied by technical auditing for "black box"-like proprietary systems. The convenience offered by conversational AI agents should be balanced against the preservation of personal data sovereignty.

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Author contributions

A.B. conceived and designed the study, formulated the research questions, developed the review protocol and search strategy, conducted the literature search, screening, and study selection, extracted and curated the data, performed the critical appraisal and conflict-of-evidence analysis, developed the taxonomy of privacy risks and mitigation approaches, analyzed the evidence on data leakage, memorization, prompt injection, inference, re-identification, surveillance, profiling, and regulatory governance, examined the risks associated with data deletion and machine unlearning, developed the integrated PPDP-based mitigation framework, prepared the tables and figures, interpreted the findings, wrote the original manuscript, and served as the corresponding author. I.K. contributed to literature screening, data extraction, verification of the synthesized evidence, and manuscript review and editing. E.E. contributed to the study design, identification of research gaps and future research directions, and critical revision of the manuscript. S.C. contributed to evidence interpretation, visualization, language editing, and manuscript review. A.A. contributed to the methodological framework, validation of the literature synthesis and critical assessment of technical privacy safeguards,

and critical review of the manuscript. A.E. contributed to the interpretation of environmental, technological, and governance aspects of the review and critically reviewed the manuscript. A.D. contributed to the conceptualization and interpretation of the findings, particularly regarding privacy-preserving approaches and governance implications. All authors read and approved the final manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable. This is a literature review; no human or animal subjects were involved. This study did not involve human participants.

Consent for publication
Not applicable. No individual person's data is included.

Competing interests
The authors declare no competing interests.

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